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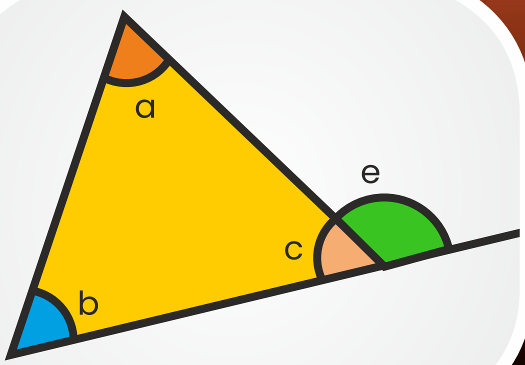

Study Material

Solution of Triangle

(English Medium)

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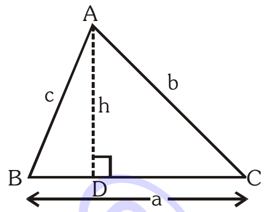
SOLUTIONS OF TRIANGLE

The process of calculating the sides and angles of triangle using given information is called solution of triangle.

In a ΔABC, the angles are denoted by capital letters A, B and C and the length of the sides opposite these angle are denoted by small letter a, b and c respectively.

1. SINE FORMULAE :

In any triangle ABC



$$\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}=\lambda=\frac{abc}{2\Delta}=2R$$

where R is circumradius and Δ is area of triangle.

Illustration 1 : Angles of a triangle are in 4 : 1 : 1 ratio. The ratio between its greatest side and perimeter is

(A) $\frac{3}{2+\sqrt{3}}$ (B) $\frac{\sqrt{3}}{2+\sqrt{3}}$ (C) $\frac{\sqrt{3}}{2-\sqrt{3}}$ (D) $\frac{1}{2+\sqrt{3}}$

Solution : Angles are in ratio 4 : 1 : 1.

$\Rightarrow$ angles are 120°, 30°, 30°.

If sides opposite to these angles are a, b, c respectively, then a will be the greatest side.

Now from sine formula $\frac{a}{\sin 120°}=\frac{b}{\sin 30°}=\frac{c}{\sin 30°}$

$$\Rightarrow \frac{a}{\sqrt{3}/2}=\frac{b}{1/2}=\frac{c}{1/2}$$

$$\Rightarrow \frac{a}{\sqrt{3}}=\frac{b}{1}=\frac{c}{1}=k \text{ (say)}$$

then $a=\sqrt{3}k$, perimeter $=(2+\sqrt{3})k$

$\therefore$ required ratio $=\frac{\sqrt{3}k}{(2+\sqrt{3})k}=\frac{\sqrt{3}}{2+\sqrt{3}}$ **Ans. (B)**

Illustration 2 : In triangle ABC, if b = 3, c = 4 and $\angle B = \pi/3$, then number of such triangles is -

(A) 1 (B) 2 (C) 0 (D) infinite

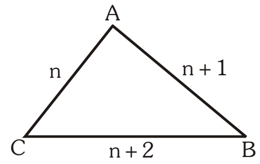
Solution : Using sine formulae $\frac{\sin B}{b}=\frac{\sin C}{c}$

$$\Rightarrow \frac{\sin \pi/3}{3}=\frac{\sin C}{4} \Rightarrow \frac{\sqrt{3}}{6}=\frac{\sin C}{4} \Rightarrow \sin C=\frac{2}{\sqrt{3}}>1 \text{ which is not possible.}$$

Hence there exist no triangle with given elements. **Ans. (C)**

Illustration 3 : The sides of a triangle are three consecutive natural numbers and its largest angle is twice the smallest one. Determine the sides of the triangle.

Solution : Let the sides be n, n + 1, n + 2 cms.



i.e. AC = n, AB = n + 1, BC = n + 2

Smallest angle is B and largest one is A.

Here, $\angle A = 2\angle B$

Also, $\angle A + \angle B + \angle C = 180°$

$\Rightarrow \quad 3\angle B + \angle C = 180° \quad \Rightarrow \angle C = 180° - 3\angle B$

We have, sine law as,

$$\frac{\sin A}{n+2} = \frac{\sin B}{n} = \frac{\sin C}{n+1} \quad \Rightarrow \quad \frac{\sin 2B}{n+2} = \frac{\sin B}{n} = \frac{\sin(180-3B)}{n+1}$$

$$\Rightarrow \quad \underset{\text{(i)}}{\frac{\sin 2B}{n+2}} = \underset{\text{(ii)}}{\frac{\sin B}{n}} = \underset{\text{(iii)}}{\frac{\sin 3B}{n+1}}$$

from (i) and (ii);

$$\frac{2\sin B\cos B}{n+2} = \frac{\sin B}{n} \quad \Rightarrow \quad \cos B = \frac{n+2}{2n} \qquad \text{......... (iv)}$$

and from (ii) and (iii);

$$\frac{\sin B}{n} = \frac{3\sin B - 4\sin^3 B}{n+1} \quad \Rightarrow \quad \frac{\sin B}{n} = \frac{\sin B(3-4\sin^2 B)}{n+1}$$

$$\Rightarrow \quad \frac{n+1}{n} = 3 - 4(1-\cos^2 B) \qquad \text{.......... (v)}$$

from (iv) and (v), we get

$$\frac{n+1}{n} = -1 + 4\left(\frac{n+2}{2n}\right)^2 \quad \Rightarrow \quad \frac{n+1}{n} + 1 = \left(\frac{n^2+4n+4}{n^2}\right)$$

$$\Rightarrow \quad \frac{2n+1}{n} = \frac{n^2+4n+4}{n^2} \quad \Rightarrow \quad 2n^2 + n = n^2 + 4n + 4$$

$$\Rightarrow \quad n^2 - 3n - 4 = 0 \quad \Rightarrow \quad (n-4)(n+1) = 0$$

$n = 4$ or -1

where $n \neq -1$

$\therefore \quad n = 4$. Hence the sides are 4, 5, 6 **Ans.**

Do yourself - 1 :

(i) If in a ΔABC, $\angle A = \frac{\pi}{6}$ and $b : c = 2 : \sqrt{3}$, find $\angle B$.

(ii) Show that, in any ΔABC : $a \sin(B - C) + b \sin(C - A) + c \sin(A - B) = 0$.

(iii) If in a ΔABC, $\frac{\sin A}{\sin C} = \frac{\sin(A-B)}{\sin(B-C)}$, show that a^2, b^2, c^2 are in A.P.

(iv) Prove that $(b-c)\cos\frac{A}{2} = a\sin\left(\frac{B-C}{2}\right)$.

(v) Prove that $(b^2 - c^2)\cot A + (c^2 - a^2)\cot B + (a^2 - b^2)\cot C = 0$.

(vi) Prove that $b^2\cos 2A - a^2\cos 2B = b^2 - a^2$.

(vii) The perimeter of a triangle ABC is six times the arithmetic mean of the sines of its angles. If the side b is equal to $\sqrt{3}$, then find angle B.

(viii) In any triangle ABC, if $\frac{a^2 - b^2}{a^2 + b^2} = \frac{\sin(A-B)}{\sin(A+B)}$, then prove that the triangle is either right angled or isosceles.

(ix) ABC is a triangle such that $\sin(2A + B) = \sin(C - A) = -\sin(B + 2C) = \frac{1}{2}$. If A, B, C are in A.P. Find A, B, C.

(x) If the angles of ΔABC are in the ratio 1 : 2 : 3, then find the ratio of their corresponding sides

(xi) In a ΔABC prove that $\frac{c}{a-b} = \frac{\tan\frac{A}{2} + \tan\frac{B}{2}}{\tan\frac{A}{2} - \tan\frac{B}{2}}$

(xii) Find the real value of x such that $x^2 + 2x$, $2x + 3$ and $x^2 + 3x + 8$ are lengths of the sides of a triangle.

2. COSINE FORMULAE :

(a) $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$ **(b)** $\cos B = \frac{c^2 + a^2 - b^2}{2ca}$ **(c)** $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$

or $a^2 = b^2 + c^2 - 2bc \cos A$

Illustration 4 : In a triangle ABC, if $B = 30°$ and $c = \sqrt{3}\,b$, then A can be equal to -

(A) 45° (B) 60° (C) 90° (D) 120°

Solution : We have $\cos B = \dfrac{c^2+a^2-b^2}{2ca} \Rightarrow \dfrac{\sqrt{3}}{2} = \dfrac{3b^2+a^2-b^2}{2\times\sqrt{3}b\times a}$

$\Rightarrow$ $a^2 - 3ab + 2b^2 = 0 \Rightarrow (a-2b)(a-b) = 0$

$\Rightarrow$ Either $a = b \Rightarrow A = 30°$

or $a = 2b \Rightarrow a^2 = 4b^2 = b^2 + c^2 \Rightarrow A = 90°$. **Ans. (C)**

Illustration 5 : In a triangle ABC, $(a^2 - b^2 - c^2)\tan A + (a^2 - b^2 + c^2)\tan B$ is equal to -

(A) $(a^2 + b^2 - c^2)\tan C$ (B) $(a^2 + b^2 + c^2)\tan C$

(C) $(b^2 + c^2 - a^2)\tan C$ (D) none of these

Solution : Using cosine law :

The given expression is equal to $-2\,bc\cos A\tan A + 2\,ac\cos B\tan B$

$= 2abc\left(-\dfrac{\sin A}{a} + \dfrac{\sin B}{b}\right) = 0$ **Ans. (D)**

Do yourself - 2 :

(i) If $a : b : c = 4 : 5 : 6$, then show that $\angle C = 2\angle A$.

(ii) In any ΔABC, prove that :

(a) $\dfrac{\cos A}{a} + \dfrac{\cos B}{b} + \dfrac{\cos C}{c} = \dfrac{a^2+b^2+c^2}{2abc}$

(b) $\dfrac{b^2}{a}\cos A + \dfrac{c^2}{b}\cos B + \dfrac{a^2}{c}\cos C = \dfrac{a^4+b^4+c^4}{2abc}$

(iii) In ΔABC, prove that :

(a) $\dfrac{\cos B}{\cos C} = \dfrac{c - b\cos A}{b - c\cos A}$ (b) $\sum \dfrac{\cos A}{c\cos B + b\cos C} = \dfrac{a^2+b^2+c^2}{2abc}$

(iv) In ΔABC, if A,B,C are in A.P., then prove that $2\cos\dfrac{A-C}{2} = \dfrac{a+c}{\sqrt{a^2+c^2-ac}}$.

(v) If in a triangle ABC, $\angle C = 60°$, then prove that $\dfrac{1}{a+c} + \dfrac{1}{b+c} = \dfrac{3}{a+b+c}$.

(vi) Triangle ABC has BC = 1 and AC = 2. Find the maximum possible value of angle A.

(vii) In a ΔABC, prove that $b^2\sin 2C + c^2\sin 2B = 2bc\sin A$.

(viii) In a ΔABC, if $9a^2 + 9b^2 = 19c^2$, then find the value of $\dfrac{\cot C}{\cot A + \cot B}$.

(ix) In a triangle ABC, the sides a, b, c are roots of $x^3 - 11x^2 + 38x - 40 = 0$. If $\sum\left(\dfrac{\cos A}{a}\right) = \dfrac{p}{q}$, then find the least value of $(p + q)$ where $p,q \in N$.

(x) In a triangle ABC, $a : b : c = 4 : 5 : 6$. Then $3A + B$ equals to

(A) 4C (B) 2π (C) $\pi - C$ (D) π

(xi) If in a triangle ABC ; sinA, sinB & sinC are in A.P. (C > 90º), then cosA is

(A) $\dfrac{3c-4b}{2b}$ (B) $\dfrac{3c-4b}{2c}$ (C) $\dfrac{4c-3b}{2b}$ (D) $\dfrac{4c-3b}{2c}$

3. PROJECTION FORMULAE :

(a) $b \cos C + c \cos B = a$ **(b)** $c \cos A + a \cos C = b$ **(c)** $a \cos B + b \cos A = c$

Illustration 6 : In a $\Delta ABC, c\cos^2\frac{A}{2} + a\cos^2\frac{C}{2} = \frac{3b}{2}$, then show a, b, c are in A.P.

Solution : Here, $\frac{c}{2}(1+\cos A) + \frac{a}{2}(1+\cos C) = \frac{3b}{2}$

$\Rightarrow$ $a + c + (c \cos A + a \cos C) = 3b$

$\Rightarrow$ $a + c + b = 3b$ {using projection formula}

$\Rightarrow$ $a + c = 2b$

which shows a, b, c are in A.P.

Do yourself - 3 :

(i) In a ΔABC, if $\angle A = \frac{\pi}{4}$, $\angle B = \frac{5\pi}{12}$, show that $a + c\sqrt{2} = 2b$.

(ii) In a ΔABC, prove that: (a) $b(a\cos C - c\cos A) = a^2 - c^2$ (b) $2\left(b\cos^2\frac{C}{2} + c\cos^2\frac{B}{2}\right) = a + b + c$.

(iii) In ΔABC, prove that $\frac{\cos A + \cos C}{a+c} + \frac{\cos B}{b} = \frac{1}{b}$.

(iv) In a ΔABC, prove that :

(a) $a(\cos B + \cos C) = 2(b + c) \sin^2 A/2$

(b) $a(\cos C - \cos B) = 2(b - c) \cos^2 A/2$

(v) In ΔABC, prove that $a(b^2 + c^2) \cos A + b(c^2 + a^2)\cos B + c(a^2 + b^2) \cos C = 3abc$.

(vi) In a ΔABC, prove that $\frac{\cos A}{c\cos B + b\cos C} + \frac{\cos B}{a\cos C + c\cos A} + \frac{\cos C}{a\cos B + b\cos A} = \frac{a^2+b^2+c^2}{2abc}$.

(vii) In ΔABC, prove that $(b + c) \cos A + (c + a)\cos B + (a + b) \cos C = a + b + c$.

4. NAPIER'S ANALOGY (TANGENT RULE) :

(a) $\tan\left(\frac{B-C}{2}\right) = \frac{b-c}{b+c}\cot\frac{A}{2}$ **(b)** $\tan\left(\frac{C-A}{2}\right) = \frac{c-a}{c+a}\cot\frac{B}{2}$ **(c)** $\tan\left(\frac{A-B}{2}\right) = \frac{a-b}{a+b}\cot\frac{C}{2}$

Illustration 7 : In a ΔABC, the tangent of half the difference of two angles is one-third the tangent of half the sum of the angles. Determine the ratio of the sides opposite to the angles.

Solution : Here, $\tan\left(\frac{A-B}{2}\right)=\frac{1}{3}\tan\left(\frac{A+B}{2}\right)$ .... (i)

using Napier's analogy, $\tan\left(\frac{A-B}{2}\right)=\frac{a-b}{a+b}.\cot\left(\frac{C}{2}\right)$ .... (ii)

from (i) & (ii) ;

$$\frac{1}{3}\tan\left(\frac{A+B}{2}\right)=\frac{a-b}{a+b}.\cot\left(\frac{C}{2}\right) \Rightarrow \frac{1}{3}\cot\left(\frac{C}{2}\right)=\frac{a-b}{a+b}.\cot\left(\frac{C}{2}\right)$$

$\{$as $A+B+C=\pi$ $\therefore \tan\left(\frac{B+C}{2}\right)=\tan\left(\frac{\pi}{2}-\frac{C}{2}\right)=\cot\frac{C}{2}\}$

$\Rightarrow \frac{a-b}{a+b}=\frac{1}{3}$ or $3a-3b=a+b$

$2a=4b$ or $\frac{a}{b}=\frac{2}{1}\Rightarrow\frac{b}{a}=\frac{1}{2}$

Thus the ratio of the sides opposite to the angles is b : a = 1 : 2. **Ans.**

Do yourself - 4 :

(i) In any ΔABC, prove that $\frac{b-c}{b+c}=\frac{\tan\left(\frac{B-C}{2}\right)}{\tan\left(\frac{B+C}{2}\right)}$.

(ii) If ΔABC is right angled at C, prove that : (a) $\tan\frac{A}{2}=\sqrt{\frac{c-b}{c+b}}$ (b) $\sin(A-B)=\frac{a^2-b^2}{a^2+b^2}$.

(iii) If in a ΔABC, $b=3c$, then find the value of $\cot\frac{A}{2}.\cot\frac{B-C}{2}$.

(iv) In a triangle ABC, $\angle A=60^\circ$ and $b:c=\sqrt{3}+1:2$, then find the value of $(\angle B-\angle C)$.

(v) ΔABC with $\angle A$ acute, $b=2$ and $c=\left(\sqrt{3}-1\right)$ has area $\left(\sqrt{3}-1\right)/2$, then find the measures of angles of triangle.

(vi) In a ΔABC if $b=3$, $c=5$ and $\cos(B-C)=\frac{7}{25}$, then find the value of $\sin\frac{A}{2}$.

(vii) Find the unknown elements of the ΔABC in which $a=\sqrt{3}+1, b=\sqrt{3}-1, C=90^\circ$.

5. HALF ANGLE FORMULAE :

$s=\frac{a+b+c}{2}$ = semi-perimeter of triangle.

(a) (i) $\sin\frac{A}{2}=\sqrt{\frac{(s-b)(s-c)}{bc}}$ (ii) $\sin\frac{B}{2}=\sqrt{\frac{(s-c)(s-a)}{ca}}$ (iii) $\sin\frac{C}{2}=\sqrt{\frac{(s-a)(s-b)}{ab}}$

(b) (i) $\cos\frac{A}{2}=\sqrt{\frac{s(s-a)}{bc}}$ (ii) $\cos\frac{B}{2}=\sqrt{\frac{s(s-b)}{ca}}$ (iii) $\cos\frac{C}{2}=\sqrt{\frac{s(s-c)}{ab}}$

(c) (i) $\tan\frac{A}{2}=\sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$ $=\frac{\Delta}{s(s-a)}$ (ii) $\tan\frac{B}{2}=\sqrt{\frac{(s-c)(s-a)}{s(s-b)}}$ $=\frac{\Delta}{s(s-b)}$ (iii) $\tan\frac{C}{2}=\sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$ $=\frac{\Delta}{s(s-c)}$

(d) Area of Triangle

$$\Delta=\sqrt{s(s-a)(s-b)(s-c)}=\frac{1}{2}bc\sin A=\frac{1}{2}ca\sin B=\frac{1}{2}ab\sin C=\frac{1}{2}ap_1=\frac{1}{2}bp_2=\frac{1}{2}cp_3,$$

where p_1,p_2,p_3 are altitudes from vertices A,B,C respectively.

Illustration 8 : If in a triangle ABC, CD is the angle bisector of the angle ACB, then CD is equal to-

(A) $\frac{a+b}{2ab}\cos\frac{C}{2}$ (B) $\frac{2ab}{a+b}\sin\frac{C}{2}$ (C) $\frac{2ab}{a+b}\cos\frac{C}{2}$ (D) $\frac{b\sin\angle DAC}{\sin(B+C/2)}$

Solution : $\Delta CAB=\Delta CAD+\Delta CDB$

$\Rightarrow \frac{1}{2}ab\sin C=\frac{1}{2}b.CD.\sin\left(\frac{C}{2}\right)+\frac{1}{2}a.CD\sin\left(\frac{C}{2}\right)$

$\Rightarrow CD(a+b)\sin\left(\frac{C}{2}\right)=ab\left(2\sin\left(\frac{C}{2}\right)\cos\left(\frac{C}{2}\right)\right)$

So $CD=\frac{2ab\cos(C/2)}{(a+b)}$

and in ΔCAD, $\frac{CD}{\sin\angle DAC}=\frac{b}{\sin\angle CDA}$ (by sine rule)

$\Rightarrow CD=\frac{b\sin\angle DAC}{\sin(B+C/2)}$ **Ans. (C,D)**

Illustration 9 : If Δ is the area and 2s the sum of the sides of a triangle, then show $\Delta\le\frac{s^2}{3\sqrt{3}}$.

Solution : We have, $2s=a+b+c$, $\Delta^2=s(s-a)(s-b)(s-c)$

Now, A.M. $\ge$ G.M.

$$\frac{(s-a)+(s-b)+(s-c)}{3}\ge\{(s-a)(s-b)(s-c)\}^{1/3}$$

or $\frac{3s-2s}{3}\ge\left(\frac{\Delta^2}{s}\right)^{1/3}$

or $\frac{s}{3}\ge\left(\frac{\Delta^2}{s}\right)^{1/3}$

or $\frac{\Delta^2}{s}\le\frac{s^3}{27}\Rightarrow\Delta\le\frac{s^2}{3\sqrt{3}}$ **Ans.**

Do yourself - 5 :

(i) Given a = 6, b = 8, c = 10. Find

(a) $\sin A$ (b) $\tan A$ (c) $\sin\frac{A}{2}$ (d) $\cos\frac{A}{2}$ (e) $\tan\frac{A}{2}$ (f) Δ

(ii) Prove that in any ΔABC, $(abcs)\ \sin\frac{A}{2}.\sin\frac{B}{2}.\sin\frac{C}{2} = \Delta^2$.

(iii) If the cotangents of half the angles of a triangle are in A.P., then prove that the sides are in A.P.

(iv) If in a triangle ABC, $\Delta = a^2 - (b-c)^2$, then find the value of $\tan A$.

(v) In any triangle ABC, prove that $(a+b+c)\left(\tan\frac{A}{2}+\tan\frac{B}{2}\right) = 2c\cot\frac{C}{2}$.

(vi) In a triangle ABC, if $\tan\frac{A}{2}.\tan\frac{B}{2} = \frac{1}{3}$ and $ab = 4$, then find the least value of c.

(vii) In ΔABC, if a,b,c (taken in that order) are in A.P. then find the value of $\cot\frac{A}{2}\ \cot\frac{C}{2}$.

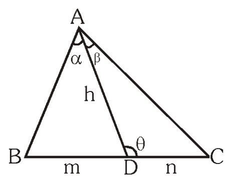
(viii) If A,B,C are the angle of a triangle, then prove that $\cot\frac{A}{2}+\cot\frac{B}{2}+\cot\frac{C}{2} = \frac{s^2}{\Delta}$

(ix) In a ΔABC if $b\sin C(b\cos C + c\cos B) = 64$, then find the area of the ΔABC.

6. m-n THEOREM :

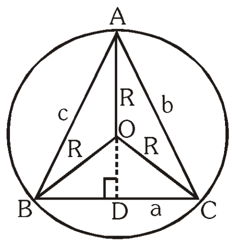
$(m+n)\cot\theta = m\cot\alpha - n\cot\beta$

$(m+n)\cot\theta = n\cot B - m\cot C.$



7. RADIUS OF THE CIRCUMCIRCLE 'R' :

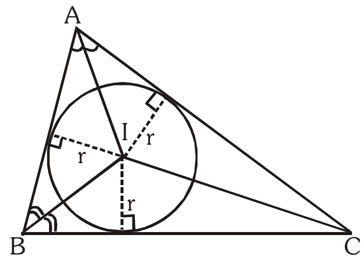
Circumcentre is the point of intersection of perpendicular bisectors of the sides and distance between circumcentre & vertex of triangle is called circumradius 'R'.



$$R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C} = \frac{abc}{4\Delta}.$$

8. RADIUS OF THE INCIRCLE 'r' :

Point of intersection of internal angle bisectors is incentre and perpendicular distance of incentre from any side is called inradius 'r'.



$$r = \frac{\Delta}{s} = (s-a)\tan\frac{A}{2} = (s-b)\tan\frac{B}{2} = (s-c)\tan\frac{C}{2} = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}.$$

$$= a\frac{\sin\frac{B}{2}\sin\frac{C}{2}}{\cos\frac{A}{2}} = b\frac{\sin\frac{A}{2}\sin\frac{C}{2}}{\cos\frac{B}{2}} = c\frac{\sin\frac{B}{2}\sin\frac{A}{2}}{\cos\frac{C}{2}}$$

Illustration 10 : In a triangle ABC, if a : b : c = 4 : 5 : 6, then ratio between its circumradius and inradius is-

(A) $\frac{16}{7}$ (B) $\frac{16}{9}$ (C) $\frac{7}{16}$ (D) $\frac{11}{7}$

Solution :

$$\frac{R}{r} = \frac{abc}{4\Delta}\Big/\frac{\Delta}{s} = \frac{(abc)s}{4\Delta^2} \Rightarrow \frac{R}{r} = \frac{abc}{4(s-a)(s-b)(s-c)} \text{ ....(i)}$$

$$\because \quad a : b : c = 4 : 5 : 6 \Rightarrow \frac{a}{4} = \frac{b}{5} = \frac{c}{6} = k \text{ (say)}$$

$$\Rightarrow \quad a = 4k,\ b = 5k,\ c = 6k$$

$$\therefore \quad s = \frac{a+b+c}{2} = \frac{15k}{2},\ s-a = \frac{7k}{2},\ s-b = \frac{5k}{2},\ s-c = \frac{3k}{2}$$

using (i) in these values $\frac{R}{r} = \frac{(4k)(5k)(6k)}{4\left(\frac{7k}{2}\right)\left(\frac{5k}{2}\right)\left(\frac{3k}{2}\right)} = \frac{16}{7}$ **Ans. (A)**

Illustration 11 : If A, B, C are the angles of a triangle, prove that : cosA + cosB + cosC = $1+\frac{r}{R}$.

Solution :

$$\cos A + \cos B + \cos C = 2\cos\left(\frac{A+B}{2}\right).\cos\left(\frac{A-B}{2}\right) + \cos C$$

$$= 2\sin\frac{C}{2}.\cos\left(\frac{A-B}{2}\right) + 1 - 2\sin^2\frac{C}{2} = 1 + 2\sin\frac{C}{2}\left[\cos\left(\frac{A-B}{2}\right) - \sin\left(\frac{C}{2}\right)\right]$$

$$= 1 + 2\sin\frac{C}{2}\left[\cos\left(\frac{A-B}{2}\right) - \cos\left(\frac{A+B}{2}\right)\right] \quad \left\{\because \frac{C}{2} = 90^\circ - \left(\frac{A+B}{2}\right)\right\}$$

$$= 1 + 2\sin\frac{C}{2}.2\sin\frac{A}{2}.\sin\frac{B}{2} = 1 + 4\sin\frac{A}{2}.\sin\frac{B}{2}.\sin\frac{C}{2}$$

$$= 1 + \frac{r}{R} \qquad \{\text{as, } r = 4R \sin A/2 \,.\, \sin B/2 \,.\, \sin C/2\}$$

$$\Rightarrow \quad \cos A + \cos B + \cos C = 1 + \frac{r}{R}. \text{ Hence proved.}$$

Do yourself - 6 :

(i) If in ΔABC, a = 3, b = 4 and c = 5, find

(a) Δ (b) R (c) r

(ii) In a ΔABC, show that :

(a) $\dfrac{a^2-b^2}{c}=2R\sin(A-B)$ (b) $r\cos\dfrac{A}{2}\cos\dfrac{B}{2}\cos\dfrac{C}{2}=\dfrac{\Delta}{4R}$ (c) $a+b+c=\dfrac{abc}{2Rr}$

(iii) Let Δ & Δ' denote the areas of a Δ and that of its incircle. Prove that

$$\Delta : \Delta' = \left(\cot\frac{A}{2}.\cot\frac{B}{2}.\cot\frac{C}{2}\right):\pi$$

(iv) If the median AD of a ΔABC makes an angle $3\pi/4$ with the side BC, then evaluate $|\cot B - \cot C|$.

(v) If in a ΔABC, AB = 4 cm, AC = 8cm and $\cos\left(\dfrac{B-C}{2}\right)=\dfrac{1}{\sqrt{2}}$; then evaluate $|2\cot B - \cot C|$.

(vi) In ΔABC, if circumradius 'R' and inradius 'r' are connected by relation $R^2 - 4Rr + 8r^2 - 12r + 9 = 0$, then find semiperimeter of the ΔABC.

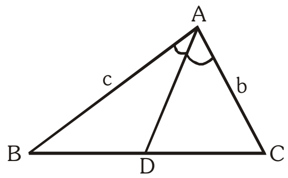
(vii) In a ΔABC, prove that $\sin 2A + \sin 2B + \sin 2C = 2\Delta/R^2$

(viii) In a ΔABC, prove that $\dfrac{b^2-c^2}{2a}=R\sin(B-C)$

(ix) In a ΔABC, prove that $a\cos B\cos C + b\cos C\cos A + c\cos A\cos B = \dfrac{\Delta}{R}$

9. ANGLE BISECTORS & MEDIANS :

An angle bisector divides the base in the ratio of corresponding sides.



$$\frac{BD}{CD}=\frac{c}{b} \Rightarrow BD=\frac{ac}{b+c} \quad \& \quad CD=\frac{ab}{b+c}$$

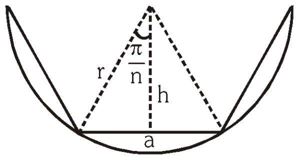
If m_a and β_a are the lengths of a median and an angle bisector from the angle A then,

$$m_a=\frac{1}{2}\sqrt{2b^2+2c^2-a^2} \text{ and } \beta_a=\frac{2bc\cos\frac{A}{2}}{b+c}$$

Note that $m_a^2+m_b^2+m_c^2=\dfrac{3}{4}(a^2+b^2+c^2)$

10. REGULAR POLYGON :

A regular polygon has all its sides equal. It may be inscribed or circumscribed.

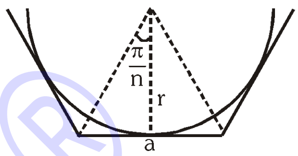


(a) Inscribed in circle of radius r :

(i) $a = 2h\tan\frac{\pi}{n} = 2r\sin\frac{\pi}{n}$

(ii) Perimeter (P) and area (A) of a regular polygon of n sides inscribed in a circle of radius r are given by $P = 2nr\sin\frac{\pi}{n}$ and $A = \frac{1}{2}nr^2\sin\frac{2\pi}{n}$

(b) Circumscribed about a circle of radius r :



(i) $a = 2r\tan\frac{\pi}{n}$

(ii) Perimeter (P) and area (A) of a regular polygon of n sides circumscribed about a given circle of radius r is given by $P = 2nr\tan\frac{\pi}{n}$ and $A = nr^2\tan\frac{\pi}{n}$

Do yourself - 7 :

(i) If the perimeter of a circle and a regular polygon of n sides are equal, then prove that $\frac{\text{area of the circle}}{\text{area of polygon}} = \frac{\tan\frac{\pi}{n}}{\frac{\pi}{n}}$.

(ii) The ratio of the area of n-sided regular polygon, circumscribed about a circle, to the area of the regular polygon of equal number of sides inscribed in the circle is 4 : 3. Find the value of n.

(iii) A regular pentagon and a regular decagon have the same perimeter, prove that their areas are in the ratio $2 : \sqrt{5}$.

(iv) If the area of circle is A_1 and area of regular pentagon inscribed in the circle is A_2, then find the ratio of areas A_1/A_2.

(v) A circle is inscribed in a regular octagon whose area is Δ. If area of a new regular octagon formed by points of contact of polygon and incircle is Δ', then $\frac{\Delta}{\Delta'}$ is

(A) $\cos^2\frac{\pi}{8}$ (B) $\sec\frac{\pi}{8}$ (C) $\sec^2\frac{\pi}{8}$ (D) $\cot^2\frac{\pi}{8}$

(vi) Let A_i, i = 1,2,3,....10 are vertices of regular decagon P. If area of polygon P is $5\sin\left(\frac{\pi}{5}\right)$ square units then the value of $(A_1A_2)(A_1A_3)(A_1A_4)....(A_1A_{10})$ is

(A) 10 (B) 9 (C) $45\sin\left(\frac{\pi}{5}\right)$ (D) $45\cos\left(\frac{\pi}{5}\right)$